

Aerodynamics-Driven Morphing Control and Flight Test for Compound Flexible Multibody Aircraft

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<https://doi.org/10.2514/1.G008194>

The compound multibody aircraft, created through flexible wingtip docking of multiple unit aircraft, not only capitalizes on high aerodynamic efficiency, but also holds the potential for in-flight reconfiguration. This study addresses the challenges of controlling the inherently unstable morphing process through aerodynamic surfaces, eliminating the need for additional driving mechanisms and offering advantages in structural simplicity. In-flight controllable morphing was achieved, significantly elevating the mission flexibility and environmental adaptability of the Compound Flexible Multibody Aircraft. A multibody dynamics model with 10 degrees of freedom is developed, and a modified control allocation strategy is proposed to enhance morphing control capability. The control structure placing the configuration controller and the overall attitude controller in parallel is developed to simultaneously guarantee the morphing and trajectory control task. Finally, a successful morphing flight test is implied, in which the configuration commands are determined according to the attainable configuration envelope. This paper presents a control method for compound flexible multibody aircraft that accommodates varying numbers of connections and provides solutions for the stable flight and aerodynamic morphing of such aircraft, expanding their usage and offering insights for the control of more general flexible aircraft.

Nomenclature

B	=	control effectiveness matrix
$C_{m\delta_{ail}}$	=	aileron-pitch coupling coefficient
$C_{m\delta_{ele}}$	=	elevator-pitch control coefficient
D	=	kinematic transformation matrix
E	=	coordinate transformation matrix
F, M, G	=	forces, moments, and gravity
H, J	=	auxiliary matrix
L	=	Lagrange function
L	=	lift
M	=	generalized mass matrix
p, q, r	=	rotational velocity
Q^*	=	generalized forces
R	=	radius vector
T, V	=	kinetic energy and potential energy
T_c	=	time constant
u, v, w	=	translational velocity
v	=	virtual control input
w, q	=	generalized velocities and displacements
w_{eq}	=	equivalent acceleration of center of mass
δ	=	control surfaces deflection
δ_{ail}^{left}	=	left aileron
δ_{ail}^{right}	=	right aileron
δ_{ele}	=	elevator
θ_{ba}, θ_{ca}	=	revolution twisting angles
ϕ_{ba}, ϕ_{ca}	=	revolution bending angles

Subscripts

a	=	unit aircraft "A"
aero	=	aerodynamic forces and moments
b	=	unit aircraft "B"
ba	=	revolution relation between unit aircraft "B" and "A"

c	=	unit aircraft "C"
ca	=	revolution relation between unit aircraft "C" and "A"
eq	=	equivalent variable
h	=	hinges
i	=	serial number of unit aircraft
$w - i$	=	from the airflow axis to the i th body axis

Superscripts

cmd	=	command value
mea	=	measured value
*	=	generalized forces or moments identifier
$\sim$	=	antisymmetric matrix operator

I. Introduction

THE concept of assembling individual aircraft into a compound configuration originated in the 1950s, however, this project [1] met its demise due to inadequate control and external disturbance. In recent years, advancements in microelectronics and evolving mission requirements [2] have rekindled researchers' interest in this field, with the goal of increasing the aspect ratio of the main wing by docking several individual aircraft, thereby enhancing aerodynamic efficiency and endurance [3,4], demonstrating the potential for high-altitude long-endurance aircraft.

Mainstream designs of the compound aircraft are wingtip-connected compound aircraft and they typically utilize movable hinge connections at the wingtips rather than fixed connections [5–14] to prevent supporting bending loads, to remove barriers associated with designing structures, and to create potential for expanding the number of connected unit aircraft. However, this design introduces an additional degree of freedom (DOF) in spanwise bending, which is inherently unstable and easily excited by disturbances [7,13,15,16].

From an optimistic perspective, loosely movable hinges allow the wingtip-connected compound aircraft to change its configuration, as shown in Fig. 1. This adaptability allows the aircraft to accommodate various flight environments. A stretched wing configuration can enhance cruising efficiency, whereas a retracted configuration offers better maneuverability, higher speed, and improved resilience to extreme atmospheric conditions. Further, based on the desire to maximize solar energy capture efficiency [17], meet specific payload requirements [2], or realize other criteria, the compound aircraft is expected to dynamically alter its configuration in flight, rather than maintaining a fixed flat configuration. Therefore, an effective

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Fig. 1 Compound flexible multibody aircraft.

configuration control method is needed to stabilize relative bending motions and maintain and intentionally alter configurations according to commands.

In this context, the term “configuration” refers specifically to the spanwise shape of the wingtip-connected compound aircraft, characterized by features such as dihedral, cathedral, or a Z-shaped bending.

Based on the aforementioned characteristics, the wingtip-connected compound aircraft is referred to as Compound Flexible Multibody Aircraft (CFMA). At first glance, its notable distinction from conventional morphing aircraft lies in three points:

1) *Morphing possibilities.* CFMA allows infinite configuration possibilities due to unrestricted connections between morphing components. Each flight unit contributes equally to the overall configuration, with no distinction between base and morphing sections. In contrast, conventional morphing aircraft only alter specific components while the base remains stationary [18–20].

2) *Morphing actuation mechanism.* The morphing is achieved by allowing each unit aircraft to autonomously alter its attitude using aerodynamic forces and moments generated by control surfaces. This method eliminates the need for additional joint actuation mechanisms [21–23], reducing system complexity and making the morphing process simpler and more flexible.

3) *Morphing duration.* The morphing process of CFMA is at a relatively slow pace and occurs throughout flight endurance, unlike systems such as AeroVironment’s Switchblade, which morph instantaneously, or those that morph intermittently during flight [18,24].

At the mechanistic level, in seeking configuration flexibility, the CFMA sacrifices the modeling- and control-friendly features of conventional morphing aircraft, leading to four main challenges in flight and configuration control:

First, the assumption of a rigid body system is invalid throughout the flight, as CFMA remains a flexible multibody system. Its flight mechanics involve coupled interactions between morphing and overall flight dynamics [10], making accurate modeling both complex and time-consuming.

Second, CFMA’s dynamic and control characteristics vary across configurations. The model describing flight and morphing can shift from stability to instability during deformation [13], whereas control surface efficiency and contributions also change. For instance, a control surface may control yaw in Configuration A but generate pitch moments in Configuration B.

Third, the maximum control capability and achievable configuration range of the aircraft despite full control effort remain unclear.

Fourth, the redundancy in control surfaces and the abundance of specific controllable quantities in a CFMA complicates control structure design and development of control methods, posing substantial engineering challenges in the deployment of prototype systems for such aircraft [25].

According to the flight profile, the CFMA should have the ability to maintain configurations and switch flexibly between different configurations, such as dihedral, cathedral, “Z,” and flat. Figuratively, this can be envisioned as a flying robot with an arm resembling an aircraft, drifting in the air.

To address these challenges, an airborne morphing and attitude control method for compound aircraft is proposed based on a multibody

flight dynamics model. To maximize morphing control, the control effectiveness matrix is reconstructed and an optimized controllable configuration envelope is obtained. The configuration control problem is then divided into overall attitude maintenance and morphing control, with independent controllers designed for each. Confident in the control effectiveness model, the Incremental Nonlinear Dynamic Inversion (INDI) method is applied to incrementally generate control commands, addressing unmodeled aerodynamic forces, modeling inaccuracies, and unknown disturbances to enhance adaptability.

In terms of flight mechanic analysis of the compound aircraft, Montalvo and Costello [12] put forward the concept of “meta-aircraft” and laid a solid foundation in perspective of flight dynamics, controllability [26], and connection control law [11]. Köthe [10] extensively explores flight dynamics employing the Kane method and potential flow theory. Their findings highlight that the stiffness and location of the joint between both wingtips play a pivotal role in the flight mechanics of connected aircraft.

In research on compound aircraft morphing control, an intuitive approach involves each unit aircraft independently controlling itself [5,8]. This strategy equips each aircraft with attitude sensors and Proportional-Integral-Derivative (PID) controllers to maintain a uniform predefined configuration (typically a flat configuration). Although easy to implement, it neglects interactions between aircraft, potentially leading to unstable motion conflicts [15]. Taking into account both morphing motion and level flight, a control structure is employed with configuration preservation as the innermost loop, attitude control as the middle loop, and trajectory tracking as the outermost loop [9,14,27]. In the configuration loop, control allocation and eigen-structure assignment are adapted to suppress spanwise deformation, equating the dynamics of compound aircraft to those of a rigid-body aircraft. In the attitude loop, conventional PID control methods [7] and state feedback [28] are applied. Montalvo et al. [29,30] validated a stability control law for a two-body articulated compound aircraft by mapping configuration deformation errors to aileron control inputs using PID control.

Nonlinear dynamic inverse (NDI) method [31,32] is a feedback linearization control technique used for handling strongly nonlinear systems. INDI is an advancement of NDI, enhancing its robustness by incrementally updating the control inputs based on the measurements of the accelerations in the assumptions that the change in control input is considered significantly faster than the change in system state [33]. INDI was initially deployed on multirotors [34], followed by tail-sitter flying wings [35,36], and transition electric Vertical Take-Off and Landing vehicles (eVTOLs). The INDI method is modeling-friendly, exhibits a certain degree of robustness, and should be suitable to be used in the morphing motion control of the CFMA, which exhibits certain degrees of instability [13].

Building on previous research, our work introduces several novel contributions: First, in-flight morphing of a compound flexible multibody aircraft using aerodynamic control surfaces was achieved for the first time. Second, to enhance morphing control capability, the control effectiveness matrix was reconstructed based on equivalent physical quantities that accurately describe CFMA’s longitudinal and lateral characteristics. Third, a compute-efficient method to solve the control allocation problem is proposed, expanding the attainable morphing envelope, which offers guidance for configuration command design. Fourth, to address configuration control and trajectory tracking simultaneously, configuration and overall attitude control should not be treated as a simple inner–outer-loop relationship. Instead, two parallel control loops with mutual feed-forward mechanisms should be developed to mitigate couplings and guarantee the operation of morphing control and overall attitude control simultaneously. This paper represents a control-oriented advancement of the previous work [13].

The paper is organized as follows: Section II introduces the proposed CFMA and its dynamics. Section III proposes the redefined control attainable set and details the improved control allocation solution method, concluding with the optimized configuration controllable envelope. Section IV provides an overview of the control structure and application of the morphing and overall attitude controller. Section V presents the flight test results.

II. Modeling

This chapter lays the foundation for dynamic analysis and control effectiveness derivation by providing a comprehensive overview of the CFMA dynamics model. To preserve analytical properties and avoid complexity, the model is based on the following assumptions:

- 1) Each unit aircraft constituting the compound aircraft is treated as an rigid body, disregarding the structural deformation.
- 2) Neglect the impact of configuration changes on the coefficients of control surface effectiveness.
- 3) Neglect the rotational friction at the hinge with proper lubrication.
- 4) Neglect the gyroscopic forces from propellers and the inertial forces when control surfaces are actuated.

5) Linear aerodynamic model is used, whereas nonlinear and unsteady aerodynamics in complex scenarios (e.g., high angles of attack, large side-slip angles, and rapid wing motion) are neglected.

Such simplifications are deemed acceptable for the subsequent INDI controller, as all unmodeled contributions to linear and angular acceleration are nonetheless considered in the control algorithm, through inertial measurement feedback.

A. Reference Frame Conventions

The compound aircraft studied in this paper is composed of three unit aircraft connected through two flexible wingtip hinges. Unit aircraft “A,” located at the center of the system, is referred to as the “reference unit aircraft,” with the body coordinate system fixed to it, as shown in Fig. 2. This configuration allows the CFMA to be considered as a treelike multibody system. The body-fixed reference frame has vectors b_x coinciding with the chord line and the wing symmetry plane of the reference unit; b_y , perpendicular to this symmetry plane; and b_z , defined to satisfy the right-hand rule.

To avoid complicating equations with coordinate transformations between the stable axes and body axes, the main wing installation angle is adjusted to match the angle of attack at the trim condition. This ensures that during level flight, vectors b_x align with the flight direction, resulting in a pitch angle of 0 deg. The direction of thrust also aligns with vectors b_x and passes through the center of gravity.

A dual-frame hinge with perpendicular rotating shafts is employed to perform two rotational DOF, as illustrated in the enlarged section in Fig. 2. The bending DOF involves rotation around the x axis of the reference unit aircraft, whereas the twisting DOF involves rotation around the y axis of unit aircraft B or C. This extends the current research [5,10], which employs single-DOF hinge along the x axis.

The bending and twisting hinges isolate the transmission of torsion and moments between unit aircraft, enabling in-flight morphing and making the rigid body assumption more reasonable. Physical constraints on bending DOF is $\pm 60^\circ$ deg, and $\pm 20^\circ$ deg in twisting DOF.

The mass and dimensions of the hinges cannot be neglected. In addition to the three unit aircraft, two rigid bodies representing the hinges should be included, resulting in a five-body, treelike, 10-DOF system.

The generalized velocities and displacements, denoted as $\mathbf{w}$ and $\mathbf{q}$, were chosen to comprehensively describe the system

$$\begin{aligned}\mathbf{w} &= [u_a, v_a, w_a, p_a, q_a, r_a, p_{ba}, p_{ca}, q_{ba}, q_{ca}]^T \\ \mathbf{q} &= [x_a, y_a, z_a, \phi_a, \theta_a, \psi_a, \phi_{ba}, \phi_{ca}, \theta_{ba}, \theta_{ca}]^T\end{aligned}\quad (1)$$

where u_a , v_a , and w_a denote the translational velocity in the body frame of the reference aircraft and p_a , q_a , and r_a denote the angular velocities. The Euler angles of the reference aircraft are ϕ_a , θ_a , and ψ_a . The variables ϕ_{ba} , ϕ_{ca} and p_{ba} , p_{ca} represent the relative bending motions, and θ_{ba} , θ_{ca} and q_{ba} , q_{ca} represent the relative twisting motions.

B. Dynamic Equations of Multibody Vehicle

Lagrange equations in quasi-coordinate form are utilized to describe the dynamics of translational, rotational, and internal relative motion in the multibody system, as shown in Eq. (2). This approach eliminates hinge constraint equations and offers clearer physical interpretation compared to Newton–Euler and standard Lagrange equations. The derivation of these equations is initially detailed in [37] and is introduced to multibody systems in the aviation context.

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{w}} \right) + H \frac{\partial L}{\partial \mathbf{w}} - J^T \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q}^* \quad (2)$$

Equation (2) can be expressed in the standard form of differential equations, denoted as Eq. (3).

$$\dot{\mathbf{w}} = M^{-1} \left(\mathbf{Q}^* - \dot{M} \mathbf{w} - H M \mathbf{w} + J^T \frac{\partial L}{\partial \mathbf{q}} \right) \quad (3)$$

with

$$\frac{\partial L}{\partial \mathbf{w}} = M \mathbf{w} \quad (4)$$

In the preceding formula, L represents the Lagrange function, defined as Eq. (5):

$$L = \sum_{i=1}^n T_i - \sum_{i=1}^n V_i \quad (5)$$

Where n is the number of components, and in this paper, n is 5. T_i and V_i denote the kinetic energy and potential energy of the i th-unit aircraft, respectively. H represents the generalized velocity matrix, with the formula as Eq. (6):

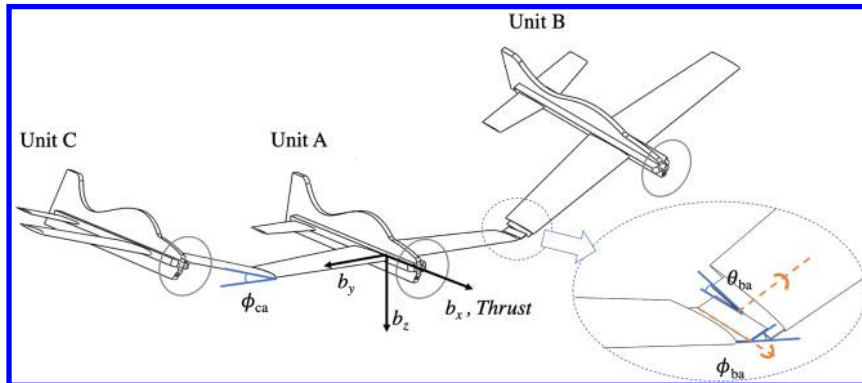


Fig. 2 Body-fixed reference frame defined in CFMA.

$$H = \begin{bmatrix} [p_a, \tilde{q}_a, r_a]^T & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} \\ [u_a, v_a, w_a]^T & [p_a, \tilde{q}_a, r_a]^T & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} & [p_{ba}, \tilde{0}, 0]^T & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & [p_{ca}, \tilde{0}, 0]^T & 0_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & [0, \tilde{q}_{ba}, 0]^T & 0_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & [0, \tilde{q}_{ca}, 0]^T \end{bmatrix} \quad (6)$$

J is a diagonal collection of transformation matrices, which can be expressed as Eq. (7), where E_{ia} is the coordinate transformation matrix from the inertial coordinate to the body coordinate of the reference vehicle. The matrix D_i transforms the Euler angle rate to the angle rate represented in the body axis

$$J = \text{diag}(E_{ia}^T, D_a^{-1}, D_{\phi_{ba}}^{-1}, D_{\phi_{ca}}^{-1}, D_{\theta_{ba}}^{-1}, D_{\theta_{ca}}^{-1}) \quad (7)$$

In this particular case, D_i can be treated as a unit matrix.

C. Generalized Forces and Moments Modeling

Q^* denotes the generalized forces and moments, expressed as Eq. (8):

$$Q^* = [F^{*T}, M^{*T}, M_1^{*T}, M_2^{*T}, M_3^{*T}, M_4^{*T}]^T \quad (8)$$

and is a function of external forces, such as aerodynamic forces, thrust, and gravity.

The expressions of the generalized forces and moments are shown in Eqs. (9–14).

Each unit of the CFMA is treated as a standard aircraft, allowing the use of a simplified aerodynamic model to categorize forces into lift, drag, pitching, rolling, and yawing moments. This approach integrates these forces into the preceding generalized-force model, avoiding the need to calculate aerodynamic forces on each wing strip as required for flexible aircraft.

$$F^* = E_{ia}(G_a + G_b + G_c + G_{h1} + G_{h2}) + F_{Ta} + E_{ah1}^T E_{h1b}^T (F_{Tb} + E_{wb} F_{Baero}) + E_{ah2}^T E_{h2c}^T (F_{Tc} + E_{wc} F_{Caero}) \quad (9)$$

$$M^* = M_a^{aero} + E_{ah1}^T E_{h1b} M_b^{aero} + E_{ah2}^T E_{h2c} M_c^{aero} + [R_{ab} + E_{ah1}^T (R_{h1} + E_{h1b} R_{bm})] \times [E_{ah1}^T E_{h1b} (E_{wb} F_b^{aero} + F_b^T)] + [R_{ac} + E_{ah2}^T (R_{h2} + E_{h2c} R_{cm})] \times [E_{ah2}^T E_{h2c} (E_{wc} F_c^{aero} + F_c^T)] + (R_{ab} + E_{ah1}^T R_{h1}) \times E_{ia} G_{h1} + (R_{ac} + E_{ah2}^T R_{h2}) \times E_{ia} G_{h2} + [R_{ab} + E_{ah1}^T R_{h1} + E_{h1b}^T R_{bm}] \times E_{ia} G_b + [R_{ac} + E_{ah2}^T R_{h2} + E_{h2c}^T R_{cm}] \times E_{ia} G_c \quad (10)$$

$$M_1^* = M_b^{aero} + [R_{h1} + E_{h1b}^T R_{bm}] \times [E_{ah1} E_{ia} G_b] + R_{h1} \times E_{ah1} E_{ia} G_{h1} + [R_{h1} + E_{h1b}^T R_{bm}] \times [E_{h1b}^T E_{wb} F_b^{aero} + F_b^T] \quad (11)$$

$$M_2^* = M_c^{aero} + [R_{h2} + E_{h2c}^T R_{cm}] \times [E_{ah2} E_{ia} G_c] + R_{h2} \times E_{ah2} E_{ia} G_{h2} + [R_{h2} + E_{h2c}^T R_{cm}] \times [E_{h2c}^T E_{wc} F_c^{aero} + F_c^T] \quad (12)$$

$$M_3^* = M_b^{aero} + R_{bm} \times [E_{h1b} E_{ah1} E_{ia} G_b] + R_{bm} \times [E_{wb} F_b^{aero} + F_b^T] \quad (13)$$

$$M_4^* = M_c^{aero} + R_{cm} \times [E_{h2c} E_{ah2} E_{ia} G_c] + R_{cm} \times [E_{wc} F_c^{aero} + F_c^T] \quad (14)$$

in which F^* and M_i^* denote the generalized forces and moments, and E_{ab} means the transform matrix from coordinate frame a to frame b . G_i and G_{hi} are the gravity of the i th-unit aircraft and i th hinges, respectively. F_i^{aero} and M_i^{aero} are the aerodynamic forces and moments of the i th-unit aircraft. The influence of hinges on aerodynamic forces can be negligibly small and thus can be ignored. R_{im} represent the vector from the center of mass of the unit aircraft to the hinge point, whereas R_{hi} denotes the dimensions of the hinge.

Given level flight conditions, aerodynamic forces and moments are calculated using a linear method, expressed as the product of static aerodynamic coefficients and dynamic pressure.

The angle of attack and airspeed vary for each unit aircraft. These are determined by correcting the reference aircraft's measured values based on the connection topology, eliminating the need for airflow sensors on each unit.

Wingtip docking affects main wing aerodynamics [38,39]. To account for this, a correction term is added to the aerodynamic coefficients of each unit aircraft, incorporating the relative dihedral angles ϕ_{ba}, ϕ_{ca} with adjacent aircraft. The definition and validation of this correction term are discussed in previous research [13].

III. Control Allocation and Configuration Envelope

This paragraph aims to improve the control allocation method with the goal of enhancing the CFMA's morphing control capabilities together with the controllable configuration envelope. In this chapter, the control effectiveness matrix was refined, and the control allocation process was optimized using an iterative dynamic weighted pseudoinverse method, improving manipulation capability while considering control surface saturation. Finally, the static controllable envelope of the configuration space was derived.

A. The Modified Control Allocation Problem

The control effectiveness matrix relates virtual control inputs v to actual command outputs δ , as shown in Eq. (15). Each aircraft is equipped with four actuators: thrust, left aileron, right aileron, and elevator. The CFMA is an overactuated system with 10 DOFs and 12 control actuators, hence its control effectiveness matrix B has fewer rows than columns; multiple outputs can result from the same input. To address this and ensure real-time on-board calculation efficiency, this research adopts the pseudoinverse method.

$$v = B\delta \quad (15)$$

The attainable moment set (AMS) quantifies control capabilities by mapping the admissible control space to moment space using the control effectiveness matrix B . It is typically visualized as a polygon or polyhedron, with its area or volume representing control capability. However, for CFMA, defining AMS is challenging due to the difficulty in establishing a uniform reference point for forces and moments. In the dynamics model, forces and moments are expressed as generalized forces, corresponding to generalized accelerations.

Therefore, to gauge CFMA's ability to change its state, the generalized accelerations under a specific control allocation method were calculated, replacing "moments" in AMS with "generalized accelerations." This modified concept is termed the attainable acceleration set (AAS).

Therefore, the control effectiveness matrix needs to be modified to accommodate the new definition. In conventional control allocation, the virtual inputs v represent desired forces or moments in the body-axis system. However, in the newly defined matrix, v corresponds to the desired generalized accelerations, as shown in Eq. (16).

$$\dot{w} = B\delta \quad (16)$$

where

$$B = M^{-1} \left(\frac{\partial Q}{\partial \delta} \right) \quad (17)$$

From Eq. (17), it is clear that the control effectiveness matrix varies with configuration, as the generalized inertial matrix M and the direction of aerodynamic forces generated by control surfaces change. The INDI method depends on accurate modeling, and using a fixed matrix during flights with multiple configurations weakens robustness. For CFMA, the control effectiveness matrix changes significantly with configuration and must be updated dynamically in each computation loop.

B. Morphing Oriented Control Effectiveness Matrix

There are 10 virtual control inputs corresponding to 10 DOFs and 12 control surfaces in the proposed CFMA. Out of the 10 DOFs, 6 belong to the rigid-body motion, whereas the remaining 4 DOFs determine the configuration of the CFMA, which is the main focus of control in this paper. All unit aircraft are expected to share the same pitching angle. Hence, θ_{ba} and θ_{ca} are desired to be controlled to 0 deg, whereas ϕ_{ba} and ϕ_{ca} dominate the configuration of the proposed CFMA and are intended to be manipulated freely. If solving a control allocation problem with all control surfaces and virtual control inputs simultaneously to ensure the controller's impact on motion is completely decoupled, it would result in severely limited configuration maneuverability, as shown in Fig. 3.

Simplifying and restructuring the control effectiveness matrix enhances the morphing maneuverability of the CFMA. This is achieved by separating overall attitude control from configuration control, as well as identifying and eliminating secondary control variables and actuators.

For a trimmed aircraft conducting level flight, significant changes in lateral translational velocity are unlikely. Therefore, $\dot{u}_a$ and $\dot{v}_a$ can be eliminated from the controlled variables.

Maintaining $\dot{w}_a = 0$ is unnecessary; instead, it is crucial to keep the vertical acceleration of the assembly's center of mass at zero. This

ensures constant potential energy during morphing, avoiding additional power consumption. The vertical acceleration of the assembly's center of mass is represented by $\dot{w}_{eq}$, replacing $\dot{w}$ as the controlled variable. In the vicinity of all three unit aircraft sharing the same mass and the same wingspan, $\dot{w}_{eq}$ can be obtained with Eq. (18):

$$\dot{w}_{eq} = \dot{w}_a + \frac{R_{bm} \cos(\phi_{ba}) \dot{p}_{ba} - R_{cm} \cos(\phi_{ca}) \dot{p}_{ca}}{3} \quad (18)$$

Further, the control effectiveness B reveals the significant gaps between configuration control variables and rigid-body attitude control variables with the fact that all ailerons contribute to $\dot{p}_a$, $\dot{p}_{ba}$, $\dot{p}_{ca}$; thrusts mainly contribute to $\dot{u}_a$ and $\dot{r}_a$, whereas all elevators affect $\dot{q}_a$ and $\dot{q}_{ba}$, $\dot{q}_{ca}$.

The overall attitude control loop was separated from the configuration control loop. Consequently, overall attitude controlled variables like $\dot{p}_a$, $\dot{q}_a$, and $\dot{r}_a$ can be omitted when considering only the configuration control problem and are considered in the overall attitude control loop. This separation does not imply a physical decoupling between overall attitude control and configuration control. Instead, it is due to the differing task priorities and control bandwidths between the two loops. In this way, the modified control effectiveness matrix links five control variables ($\dot{p}_{ba}$, $\dot{p}_{ca}$, $\dot{q}_{ba}$, $\dot{q}_{ca}$, $\dot{w}_{eq}$) with all 12 actuators.

The contribution of thrusts and elevators to deformation exists only in the existence of the relative twisting angle (θ_{ba} , θ_{ca}), which is expected to be controlled to 0 deg in this paper. Therefore, thrust, as a control signal, can be eliminated. This reduces the control effectiveness matrix to a 5×9 matrix, with five controlled variables ($\dot{p}_{ba}$, $\dot{p}_{ca}$, $\dot{q}_{ba}$, $\dot{q}_{ca}$, $\dot{w}_{eq}$) and nine manipulated variables (one elevator and two ailerons per unit aircraft).

In terms of stability, twisting dynamics are prominent among other motions, with rapid convergence characteristics similar to the short period. In terms of controllability, elevators exhibit superior control effectiveness over the twisting motion compared to other control surfaces, owing to their larger control surface area and longer moment arm. Given this insight, the control signal of the elevators and the virtual control input of the twisting angle can be excluded from the control allocation problem. The control signal commands for the elevators is shown in Eq. (19).

$$\delta_{ele_i} = \frac{C_{m\delta_{ail}}}{C_{m\delta_{ele}}} \left(\delta_{ail_i}^{\text{left}} - \delta_{ail_i}^{\text{right}} \right) + \text{PI}(\theta_{i,\text{err}}) \quad (19)$$

where the first term represents the feed-forward term used to compensate for the disturbance caused by the aileron manipulation on the pitch angular acceleration. The latter term corresponds to the cascade proportion-integrate feedback controller to eliminate the twisting angle.

The control effectiveness matrix is now reduced to the size of 3×6 , with the virtue control input list as $[\dot{p}_{ba}, \dot{p}_{ca}, \dot{w}_{eq}]^T$ and all six ailerons, significantly expanding control capabilities in the configuration space. As shown in Fig. 3, under level flight conditions, the AAS with the simplified matrix (b) shows a substantial improvement over the baseline matrix of 10×12 (a).

The improvement can be explained by a simple example, as shown in Fig. 4. The subplot (a) indicates the morphing trends guided by $B_{10 \times 12}$, whereas the subplot (b) corresponds to $B_{3 \times 6}$. In each subplot, dashed lines indicate the original configuration, and solid lines show the morphing trends. Blue arrows represent the direction of the incremental forces generated by control surface deflection. For symmetric upward bending, the $B_{10 \times 12}$ matrix lifts the side units

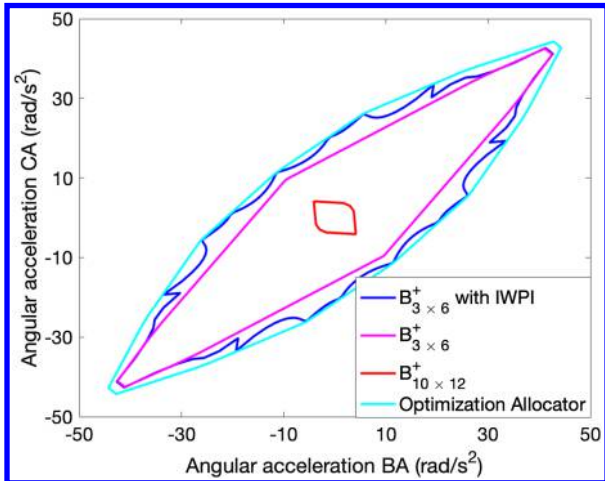


Fig. 3 Attainable generalized acceleration.

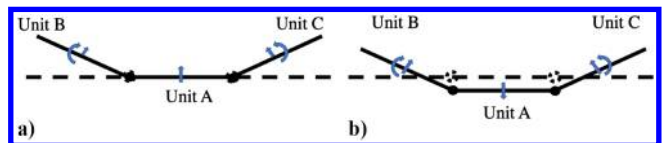


Fig. 4 Morphing trends of the CFMA under two control allocators.

while keeping the central unit still. In contrast, the simplified $B_{3 \times 6}$ matrix actively drives the central unit to descend to coordinately achieve the configuration control, avoiding increased potential energy and reducing control surface deflections.

In summary, the modified control allocator, free from rigid-body moving constraints, coordinates unit movements for morphing, providing more control authority without compromising precision.

C. The Improved Control Allocation Solving Method

Building on Subsection B, an efficient approach is sought to solve the control allocation problem, with a focus on expanding the AAS while considering actuator saturation. The pseudoinverse method provides the solution with the smallest L_2 norm, but this conflicts with the smallest L_∞ norm solution required for expanding the AAS.

Employing optimization-based methods to find the solution with minimal L_∞ norm may compromise real-time computational efficiency. The Iterative Weighted Pseudoinverse (IWPI) method was proposed to solve the control allocation problem, facilitating onboard microcontroller operations and approximating the results achieved by optimization algorithms. The approach is inspired by the PAN algorithm mentioned in [40]. However, our proposed method is more intuitive and avoids computational uncertainties.

Formulation of the IWPI is shown in Eq. (20).

$$\delta_k = W_k^{-1} (B W_k^{-1})^{-1} v \quad (20)$$

where δ_k is the k th iterative result of the control allocation problem and v is the virtue control input. W_k is the k th iterative weighting matrix and is expressed in Eq. (21).

$$W_k = \text{diag} \left(\begin{bmatrix} \sqrt{\frac{\delta_{\text{ail}_b, \text{left}}^{d, k-1}}{\delta_{k-1}^{d-\max}}}, \sqrt{\frac{\delta_{\text{ail}_b, \text{right}}^{d, k-1}}{\delta_{k-1}^{d-\max}}}, \sqrt{\frac{\delta_{\text{ail}_a, \text{left}}^{d, k-1}}{\delta_{k-1}^{d-\max}}}, \sqrt{\frac{\delta_{\text{ail}_a, \text{right}}^{d, k-1}}{\delta_{k-1}^{d-\max}}}, \\ \sqrt{\frac{\delta_{\text{ail}_c, \text{left}}^{d, k-1}}{\delta_{k-1}^{d-\max}}}, \sqrt{\frac{\delta_{\text{ail}_c, \text{right}}^{d, k-1}}{\delta_{k-1}^{d-\max}}} \end{bmatrix} \right) \quad (21)$$

where δ^d is normalized by $\delta^d = \delta / \delta_{\text{sat}}$. δ_{sat} is the saturation constraint of the control surface. W_0 is the unit matrix. $\delta^{d-\max}$ is the maximum deflection value among all aileron surfaces.

The control surface associated with a larger command in the preceding allocation is assigned a higher weight in the current allocation iteration. After multiple iterations, surfaces tend to reach saturation simultaneously. This idea is to encourage less proficient workers to contribute more to prevent overburdening highly efficient ones. Table 1 shows the control surfaces deflection under the command virtue input of $p_{\text{ba}}^{\text{cmd}} = -10 \text{ rad/s}^2$ and $p_{\text{ca}}^{\text{cmd}} = 10 \text{ rad/s}^2$. Although this algorithm can be further iterated to achieve greater performance gains, the marginal benefits diminish. Typically, setting $k = 2$ is satisfying.

Figure 3 displays the AAS envelope obtained by solving the control allocation problem using IWPI, where the area of (c) is 26.65% larger than that of (b), and 4097% larger than that of (a). Panel (d) represents the boundary obtained by minimizing the L_∞ through optimization, whereas the area of (c) is 14.18% smaller than

that of (d). However, the computational speed of IWPI remains between 981 and 1061 Hz in multiple tests, significantly surpassing that of the optimization method (6–7 Hz) in the same computational platform.

D. Controllable Configuration Envelope Definition

The previous studies [7,13] identified hazardous configurations for this aircraft, where divergence occurs despite full control input. To ensure feasible commands, defining the configuration envelope, which are configurations the CFMA can maintain within control limits, is essential.

Different configurations of CFMA correspond to varying dynamics and control effectiveness. The controllable configuration envelope is identified by seeking the extreme trimming static configuration boundaries using the Distance Fields Over Grids (DFOG) method [41,42].

The method begins by choosing a space that contains the configuration envelope R and discretizing G into a grid g . Each point in the grid is then projected onto R by finding the point within R that has the minimum Euclidean distance to the point in the grid, formulated as an optimal control problem. All grid points g outside the envelope R project onto the boundary of the envelope.

In this particular case, G is the configuration space, with the variables describing the configurations as the orthogonal basis. Here, the x axis is ϕ_{ba} , and the y axis is ϕ_{ca} . g in G represents a specific configuration. For each g , solve the optimal Problem 1 to find the projection onto R .

Problem 1: Given the grid point g and the optimal variable y , the following problem realizes the minimum distance projection.

$$\text{minimize } J = \frac{1}{2} \|y - g\|_2^2 \quad (22)$$

subject to

$$B = f_1(y, \dots), Q^* = f_2(y, \dots), M = f_3(y, \dots) \quad (23)$$

$$-\frac{Q^*}{M} \Big|_{w_{\text{eq}}, p_{\text{ba}}, p_{\text{ca}}} \leq \text{IWPI}(B_{3 \times 6}) \delta \quad (24)$$

$$\delta \in \mathbb{U} \quad (25)$$

$$u_a \leq u_{\text{max}} \quad (26)$$

The constraints are as follows: Eq. (24) implies that the potential deformation motion tendency induced by inherent dynamics can be compensated for by the control surfaces in the aforementioned control allocation method. The case ‘=’ indicates that the CFMA is trimmed at a certain configuration, without any tendency of deformation. Equation (25) represents that all control surfaces should not violate their limits. Equation (26) indicates that the CFMA does not exceed the maximum air speed.

The results of the configuration controllable envelope are illustrated in Fig. 5. It should be noted that the length of the envelope polygon in the “ $\phi_{\text{ba}} = \phi_{\text{ca}}$ ” direction is larger than that in the “ $\phi_{\text{ba}} = -\phi_{\text{ca}}$ ” direction. This implies that CFMA can operate in exaggerated Z configurations but is constrained in dihedral or cathedral configurations.

The derived envelope is relatively aggressive for the following three reasons:

1) The boundary is defined under the assumption of an ideal controller. Calculations involving delays, sensor noise, and other factors can degrade the dynamic characteristics of the controller, potentially leading to configuration states beyond the boundary.

2) The boundary is a neutral trimming configuration. The CFMA is particularly sensitive to external disturbances, and minor aerodynamic perturbations can easily cause the CFMA to exceed the region.

Table 1 Control signal under multiple allocators

Control surface /rad	$B_{10 \times 12}$	$B_{3 \times 6}$	$B_{3 \times 6}$ with IWPM
$\delta_{\text{ail}_b}^{\text{left}}$	0.2425	0.1044	0.0876
$\delta_{\text{ail}_b}^{\text{right}}$	-0.0272	-0.0208	-0.0876
$\delta_{\text{ail}_a}^{\text{left}}$	0.0908	-0.0835	-0.0876
$\delta_{\text{ail}_a}^{\text{right}}$	0.0908	-0.0835	-0.0876
$\delta_{\text{ail}_c}^{\text{left}}$	-0.0272	-0.0208	-0.0876
$\delta_{\text{ail}_c}^{\text{right}}$	0.2425	0.1044	0.0876
L_∞	0.2425	0.1044	0.0876

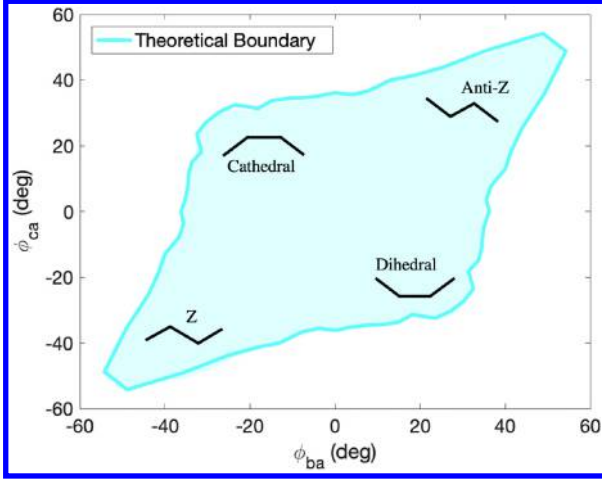


Fig. 5 Controllable configuration envelope.

3) Balancing at the boundary implies that there is no additional control margin available to perform additional maneuvers or stabilization.

Therefore, the target configuration should not be set too close to the boundary.

IV. Control Design

A. Overview of the Control Scheme

The control objective is to accurately track the configuration commands within the controllable envelope during flight, derived in Section III.D. The configuration commands are denoted as ϕ_{ba}^{cmd} and ϕ_{ca}^{cmd} , with the desired relative twist θ_{ba}^{cmd} and θ_{ca}^{cmd} set to 0.

As stated in Section III.B, due to control priority and manipulation effectiveness, the proposed control structure is divided into an overall attitude control loop and a configuration control loop, as shown in Fig. 6. This division clarifies logical relationships and effectively utilizes existing flight control methods.

The configuration controller uses all ailerons to manipulate the relative bending angles of each unit aircraft, with elevators compensating for extra pitching moments generated by the ailerons. The overall attitude controller utilizes elevators, thrust, and ailerons for rigid-body pitch, yaw, and roll control. Commands from both control loops are added and published to the actuators.

The configuration loop has higher control priority and requires more aggressive maneuvers but has weaker stability, whereas the attitude loop provides stronger controllability and stability with less precision and milder maneuvers. Therefore, the configuration loop uses an INDI controller, whereas a conventional cascade proportional-integrate (PI) controller suffices for the overall attitude control loop.

B. Configuration Controller

The configuration loop consists of several components. First, a logarithmic outer-loop controller converts configuration angle error to desired angular velocity. The INDI controller then calculates the

desired acceleration, which acts as a virtual control input for a real-time updated control allocator, as detailed in Section III.B.

The INDI controller is employed in the knowledge that control input changes are faster than system state changes, as the actuator frequency (22 rad/s) from experiments significantly differs from the highest CFMA morphing mode frequency (1.7 rad/s).

INDI enables precise control by incrementally adjusting control inputs, relying on the local input–output measurement and addressing inverted control effectiveness dynamics. This enhances robustness against unmodeled aerodynamics from irregular aircraft shapes and complex wingtip hinge airflow. Equation (27) gives the incremental control law

$$v_c = v_f + (\dot{w}_c - \dot{w}_m^f) \quad (27)$$

where v_c is the desired angular acceleration, viewed as the virtual control input for the control allocator; v_f is the virtual control input in the last calculation loop. $\dot{w}_m^f$ denotes the measured angular acceleration $\dot{p}_{ab}$ and $\dot{p}_{ac}$ smoothed by a low-pass filter. As angular acceleration is now a regulated variable, command angular rates $\dot{w}_c$ can be managed through simple proportional feedback.

$$\dot{w}_c = K_w(w_c - w_m) \quad (28)$$

The angular velocity command w_c is obtained as Eq. (29), and it can be viewed as an adaptive variable parameter controller with boundary, enabling the controller to expedite error elimination when the error is large, while allowing for gentler control inputs when the error is small.

$$w_c = \frac{K_e e_{\text{angle}}}{b^2 - e_{\text{angle}}^2} \quad (29)$$

where $e_{\text{angle}} = \phi_{ba,ca}^{cmd,f} - \phi_{ba,ca}^{mea}$; K_e and b are tuned constants.

The configuration command $\phi_{ba,ca}^{cmd,f}$ is smoothed by a low-pass filter to avoid jerk.

The final output is obtained by control allocator (30):

$$\delta_{\text{Aileron}} = \text{IWPI}(B_{3 \times 6})v_c \quad (30)$$

where the variation of control surface effectiveness with changes in aircraft configuration and flight conditions has been taken into account.

C. Overall Attitude Controller

The overall attitude controller uses a cascade PI method. Unlike conventional aircraft, it controls the attitude of the entire assembly rather than that of individual unit aircraft. The physical quantities representing the assembly's attitude are called the equivalent attitude.

The equivalent attitude $\phi_{eq}, \theta_{eq}, \psi_{eq}$ and the corresponding angular velocities p_{eq}, q_{eq}, r_{eq} are functions of the attitudes $\phi_{eq}, \theta_{eq}, \psi_{eq}$, angular velocities p_a, q_a, r_a , and relative motion $p_{ba}, p_{ca}, \phi_{ba}$, and ϕ_{ca} .

Because the constitutive units within the assembly are only expected to rotate around the x -axis, the equivalent pitch and yaw

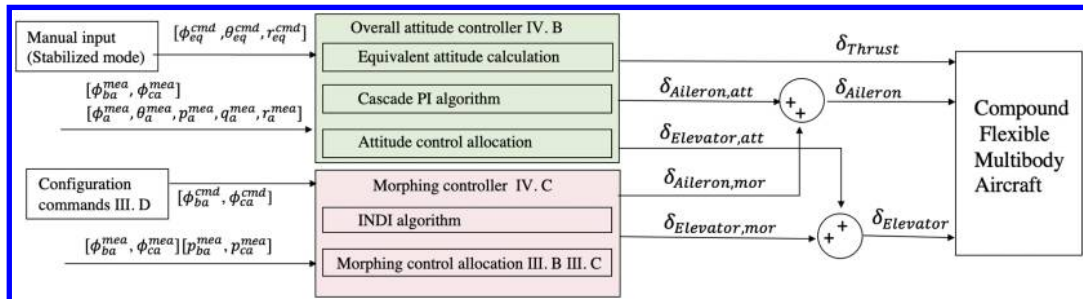


Fig. 6 Overview of the control structure.

angles are shared with the central body, whereas the equivalent rolling angle differs from that of the central body and is expressed in Eqs. (32) and (33), which is the solution of Eq. (31). Equation (31) provides an approximate estimate of the lateral force, derived from the geometric relationships in the lateral direction of aerodynamic forces.

$$\begin{aligned} & L_a \sin(\phi_a) + L_b \sin(\phi_a + \phi_{ba}) + L_c \sin(\phi_a + \phi_{ca}) \\ &= \sin(\phi_{eq})(L_a + L_b + L_c) \end{aligned} \quad (31)$$

Equation (31) provides an approximate estimate of the lateral force, derived from the geometric relationships in the lateral direction of aerodynamic forces.

$$\phi_{eq} = \arctan\left(\frac{\sin(\phi_a) + \sin(\phi_a + \phi_{ba}) + \sin(\phi_a + \phi_{ca})}{\cos(\phi_a) + \cos(\phi_a + \phi_{ba}) + \cos(\phi_a + \phi_{ca})}\right) \quad (32)$$

$$[\theta_{eq}, \psi_{eq}]^T = [\theta_a, \psi_a]^T \quad (33)$$

Equivalent angular velocities are formulated as Eq. (34):

$$w_{eq}^m = [p, q, r]_{eq}^T = E_a^{eq}[0, q_a, r_a]^T + [\dot{\phi}_{eq}, 0, 0]^T \quad (34)$$

where

$$E_a^{eq} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi_{relative}) & \sin(\phi_{relative}) \\ 0 & -\sin(\phi_{relative}) & \cos(\phi_{relative}) \end{bmatrix} \quad (35)$$

$$\phi_{relative} = \phi_{eq} - \phi_a \quad (36)$$

The desired body angular acceleration $\dot{p}_{eq}^{cmd}$, $\dot{q}_{eq}^{cmd}$, and $\dot{r}_{eq}^{cmd}$ are derived as Eq. (37):

$$\begin{aligned} \dot{p}_{eq}^{cmd} &= k_{ip}^1 \left\{ \frac{1}{T_c} (\phi_{eq}^{cmd} - \phi_{eq}^{mea}) \right\} + k_i^1 \int (\phi_{eq}^{cmd} - \phi_{eq}^{mea}) - p_{eq}^{mea} \\ \dot{q}_{eq}^{cmd} &= k_{ip}^2 \left\{ \frac{1}{T_c} (\theta_{eq}^{cmd} - \theta_{eq}^{mea}) \right\} + k_i^2 \int (\theta_{eq}^{cmd} - \theta_{eq}^{mea}) - q_{eq}^{mea} \\ \dot{r}_{eq}^{cmd} &= k_{ip}^3 \left\{ (r_{eq}^{cmd} - r_{eq}^{mea}) \right\} \end{aligned} \quad (37)$$

where k_{ip} , T_c , and k_i denote the inner-loop proportional gain, outer-loop time constant, and outer-loop integral gain, respectively.

The desired equivalent body angular acceleration are then converted to actuator signals Eq. (38) via an equivalent attitude-oriented control effectiveness matrix stated as Eq. (39).

$$u_\delta = B_{eq}^{att} [\dot{p}_{eq}^{cmd}, \dot{q}_{eq}^{cmd}, \dot{r}_{eq}^{cmd}]^T \quad (38)$$

where

$$B_{eq}^{att} = E_a^{eq} \left[M^{-1} \left(\frac{\partial Q}{\partial u} \right) \right]_{4-6rows} \quad (39)$$

V. Configuration Control Test

The CFMA concept and its controller are evaluated through morphing experiments. The target configurations include two types: symmetric (dihedral and cathedral) and asymmetric (Z and anti-Z). In symmetric configurations, the unit aircraft on both sides perform equal and opposite roll movements while the central aircraft remains level. In asymmetric configurations, one side flips upward and the other downward, resembling a straightened Z. Asymmetric commands do not specify the central unit's attitude; this is determined by the overall attitude controller based on the equivalent attitude.

A. Experiment Setup

The experiments will be conducted outdoors, with gust wind speeds maintained below 2 m/s. The prototype is shown in Fig. 7, consisting of three identical commercially available EPP aircraft connected at the wingtips. The connections between each unit aircraft exhibit significant flexibility. The weight of each unit aircraft fluctuates between 896 and 947 g, with differences in weight attributed to variations in equipment and cable lengths. The wingspan of each unit aircraft is identical at 89 cm, with a mean aerodynamic chord length of 24 cm. The hinges are three-dimensionally printed, with each weighing 102 g. The total weight of the compound aircraft is 3.0 kg, and the wingspan from tip to tip is 276 cm, with a cruising speed of 11 m/s.

The flight control algorithm is deployed on a PIXHAWK onboard hardware. The firmware is a custom version based on the PX4 open-source code. Each unit aircraft is equipped with a flight controller board. All the controller boards are connected by serial port to transfer measured data and control commands. The control algorithm mentioned in this paper operates on the flight control board of the central unit aircraft, which outputs the desired pitch angle, aileron deflections, and thrust commands for each unit aircraft. The flight control boards on other units receive these commands, drive their respective servos, and also transmit the measured attitude angles and angular velocities to the flight control board of the reference aircraft. The configuration angles (ϕ_{ba} and ϕ_{ca}) and their corresponding angular velocities ($\dot{p}_{ba}$ and $\dot{p}_{ca}$) are measured by two ways: 1) reading analog values by potentiometers mounted on the hinge, and 2) performing the conjugate multiplication of the quaternions and calculating the roll rate difference between adjacent aircraft. The data from these two sources will be fused to obtain a more accurate value. The angular accelerations are obtained by taking the derivatives of the filtered angular velocities.

B. CFMA with Three Unit Aircraft

An automatic morphing sequence is developed, divided into four stages, each involving a transition to a new configuration, holding for 4 s, then returning to the flat configuration and holding again for 4 s. The stages proceed as follows: Stage 1: cathedral configuration; Stage 2: Z configuration; Stage 3: dihedral configuration; Stage 4: anti-Z configuration.

The navigation controller was not used in the flight experiments; instead, the CFMA was operated in stabilized mode. In this mode, the pilot's joystick input directly mapped to the desired overall roll angle, pitch angle, yaw rate, and total thrust, while a switch activated the morphing sequence. The route of the flight test is shown in Fig. 8.

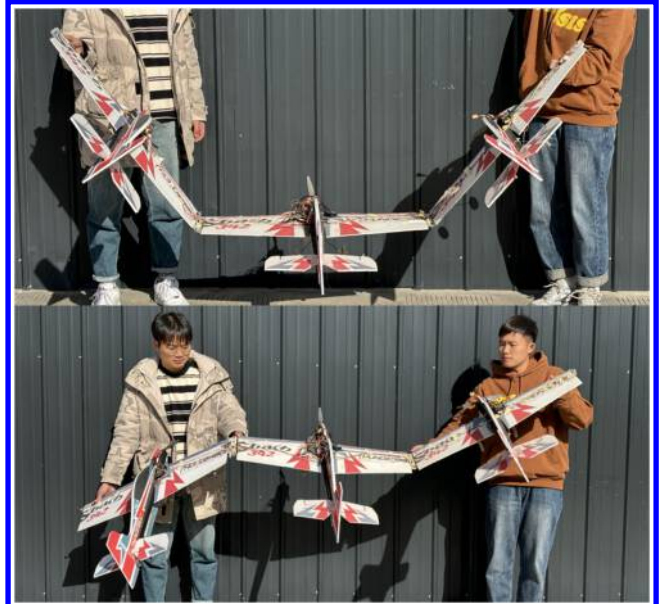


Fig. 7 The CFMA prototype.

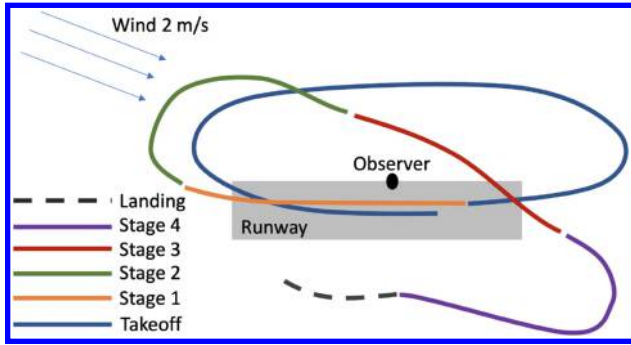


Fig. 8 Flight path in the morphing test.

Figure 9 presents ground photographs of the CFMA's folding process, with four rows of subplots displaying the entire morphing sequence. The first row corresponds to Stage 1, and so on for each subsequent row. The flight data are shown in Fig. 10, where subplot (a) displays the configuration control data, subplot (b) shows the aileron deflection, subplot (c) depicts the equivalent roll angle, and subplot (d) represents the pitch angle of all the unit aircraft. Note that the horizontal axis represents time, which starts from the morphing phase rather than takeoff.

According to the controllable configuration envelope, the CFMA provides a larger maneuvering range in the Z configuration. Therefore, the configuration command amplitude for symmetric morphing is set at 0.4 rad (22.92 deg), whereas for the Z configuration, it is increased to 0.6 rad (34.38 deg). The experimental configuration data together with the controllable configuration envelope is shown in Fig. 11. To avoid control saturation, which could lead to a failure in overall attitude control, the target configuration amplitudes were chosen to maintain a safe margin from the boundaries.

Experimental data indicates that the switching process of the CFMA configuration is rapid. However, during the maintenance phase in Stage 3 (dihedral configuration), it still exhibits tracking errors of approximately 5 deg. During the other configuration maintained, disturbed by unmeasured gust, the measured data exhibits a maximum deviation of 2 deg.

In both flat and symmetric configurations, the equivalent roll attitude aligns with the central aircraft's roll angle. However, in the Z configuration, a noticeable divergence occurs. At this point, the

overall motion trend of the aircraft does not undergo significant changes, demonstrating that the attitude of any individual-unit aircraft cannot represent the attitude of the entire CFMA. Conversely, the equivalent attitude effectively captures this representation.

The pitch angles of the three unit aircraft are nearly equal, exhibiting similar motion trends. This reflects the effectiveness of the control strategy for relative twisting angles. Furthermore, pitch angle variation shows no significant correlation with the configuration control; instead, the trend in pitch angle follows the manual input.

C. CFMA with Two Unit Aircraft

To demonstrate the broad applicability of the control method proposed in this paper, a configuration control flight test was conducted using a CFMA composed of two unit aircraft. During the transition from a three-aircraft assembly to a two-aircraft assembly, the aforementioned control allocation method, equivalent attitude definition, control structure, and all control parameters were retained. The necessary adjustments involved removing the parameters related to the third aircraft from the multibody dynamic model, recalculating the effectiveness matrix B , and eliminating specific rows and columns to derive a 2×4 matrix. In this particular case, unit aircraft C was removed.

As in the three-body CFMA test, the two-body CFMA took off and gained altitude in the first lap like a conventional aircraft. The morphing sequence was activated during both level flight and turning stages to verify that the configuration change does not impact overall attitude control.

Figure 12 presents ground photographs of the CFMA's folding process, with three rows of subplots displaying the entire morphing sequence. The corresponding flight data are shown in Fig. 13, where subplot (a) displays the configuration control data, subplot (b) shows the aileron deflection, subplot (c) depicts the equivalent roll angle, and subplot (d) represents the pitch angle of all the unit aircraft.

The aircraft initially transitions from a flat configuration to a 0.4-rad (22.92-deg) cathedral configuration, maintains this setup for 22 s, and then promptly switches to a 0.5-rad (28.65-deg) dihedral configuration. After holding for 32 s, it reverts to the flat configuration.

Overall, the configuration control response is rapid with minimal fluctuations. No overshoot is observed across all configuration transitions.

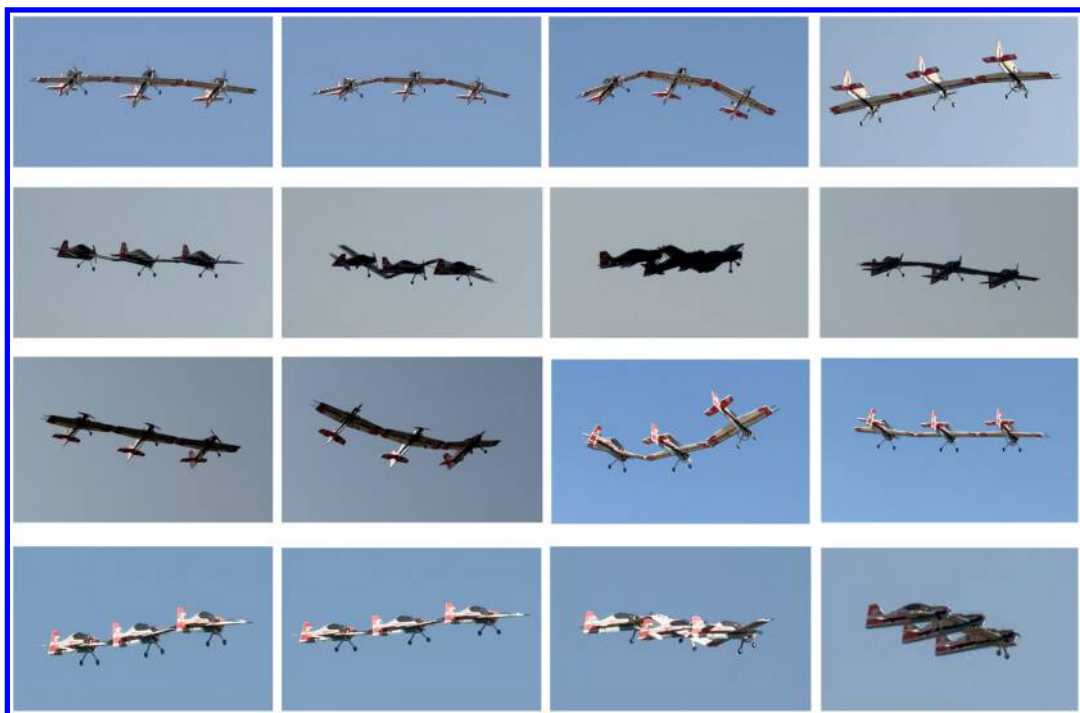


Fig. 9 Photographs of the morphing process.

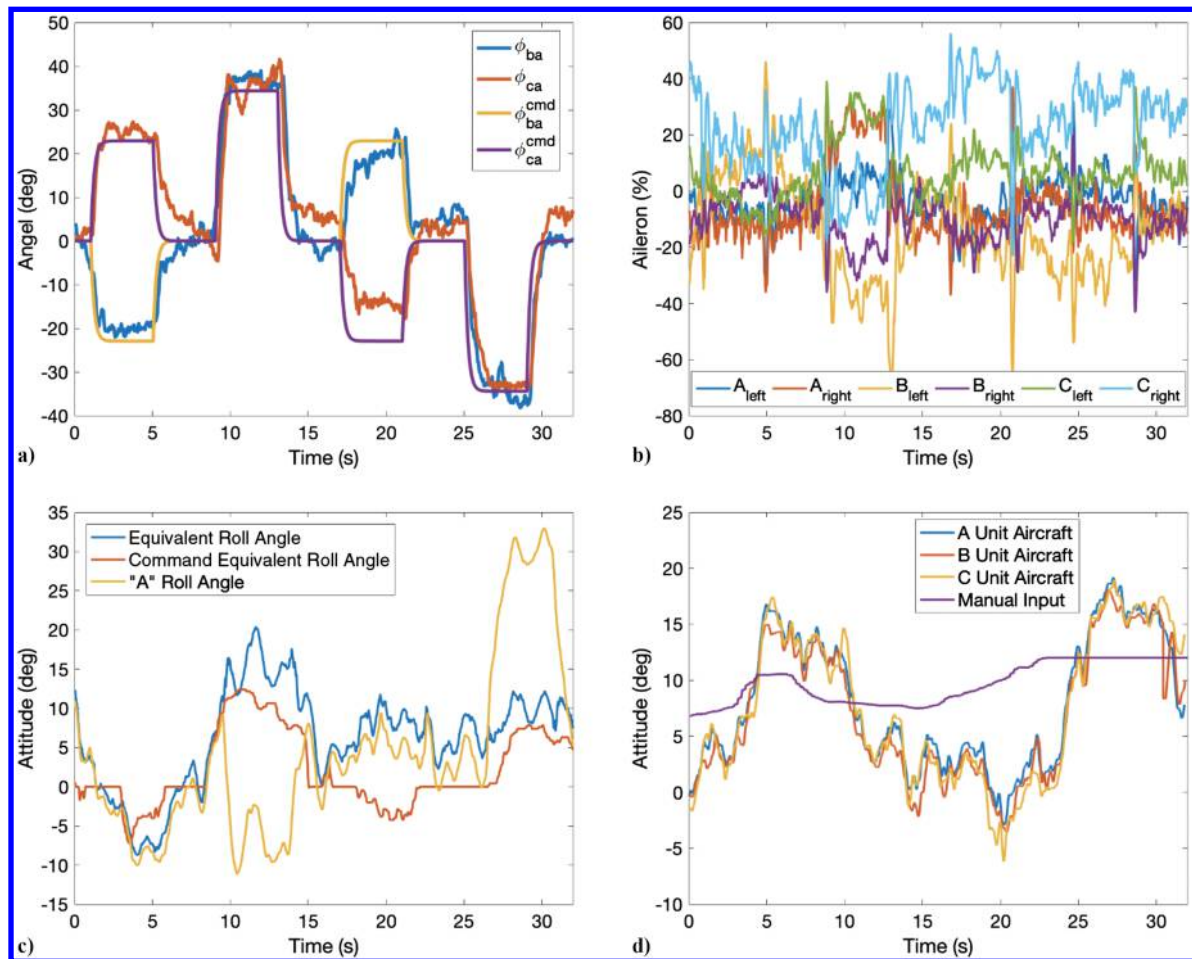


Fig. 10 Data on the morphing experiment.

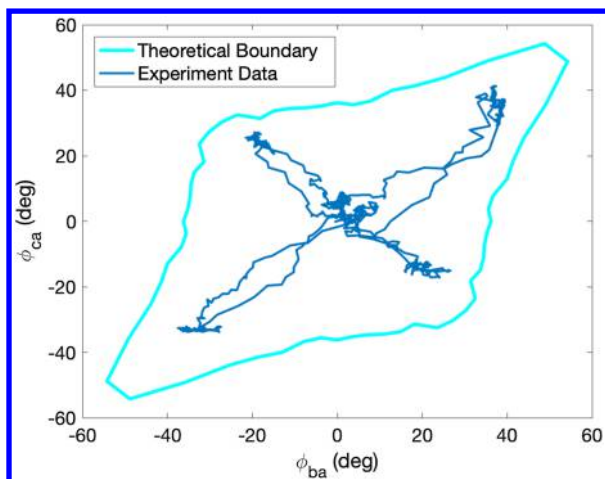


Fig. 11 Configurations in the controllable envelope.

D. Discussion

The preceding experiments reveal certain commonalities, and the derivation is general.

The configuration remains stable even in atmospheric turbulence. The morphing process is completed quickly, with limited fluctuations, demonstrating satisfactory response speed and control precision.

The pitch attitude is not affected by the morphing motion, as the overall attitude controller automatically manages attitude control and responds to the pilot's inputs. The pitch angles of all the unit aircraft are similar. The entire CFMA does not significantly differ from that of

a standard aircraft with a flat configuration. Both the configuration control task and the manual trajectory control task are executed simultaneously.

The definition of equivalent attitude is meaningful. In a flat configuration, the equivalent roll angle coincides with the roll angle of the unit aircraft, but they diverge significantly when the configuration changes. Introducing the equivalent roll angle to replace the roll angle of the unit aircraft prevents the attitude controller from being confused by the variations in the roll angle of any specific aircraft during configuration changes.

Discrepancies between experimental results and commands have been observed. Excluding the effects of wind interference, the main technical causes are that the low-cost inertial sensors may encounter challenges in providing precise measurements, and the elastic deformation of the unit aircraft introduces phase lag among all the distributed sensors. These two factors result in variations and drift in the measured pitch, roll angles, and roll rates. Other causes, such as unmodeled complex aerodynamic flow at the hinges and manufacturing discrepancies in mass and inertia among each unit aircraft, also adversely affect control performance.

The feasibility of configuration maintenance in a CFMA with both odd and even numbers of unit aircraft has been experimentally demonstrated. In the future, we will augment the number of connection units and facilitate more intricate configuration deformations.

VI. Conclusions

An investigation into the configuration holding and in-flight morphing control approach, and a successful morphing flight test with a satisfying performance was conducted. Aerodynamic control surfaces are employed to drive the morphing process, eliminating the need for actuator-driven mechanisms.

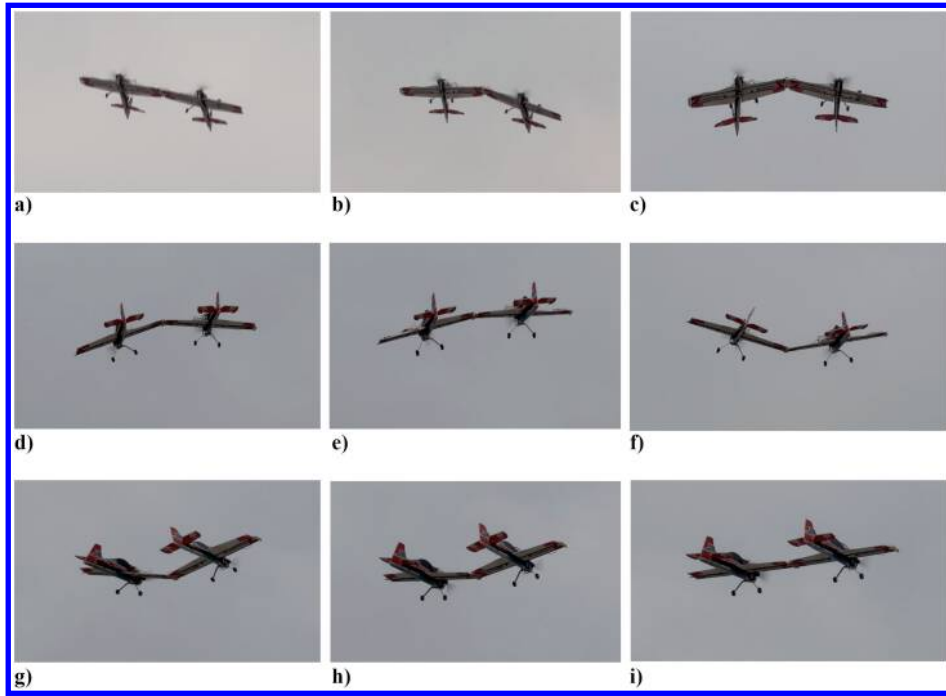


Fig. 12 Photographs of the morphing process.

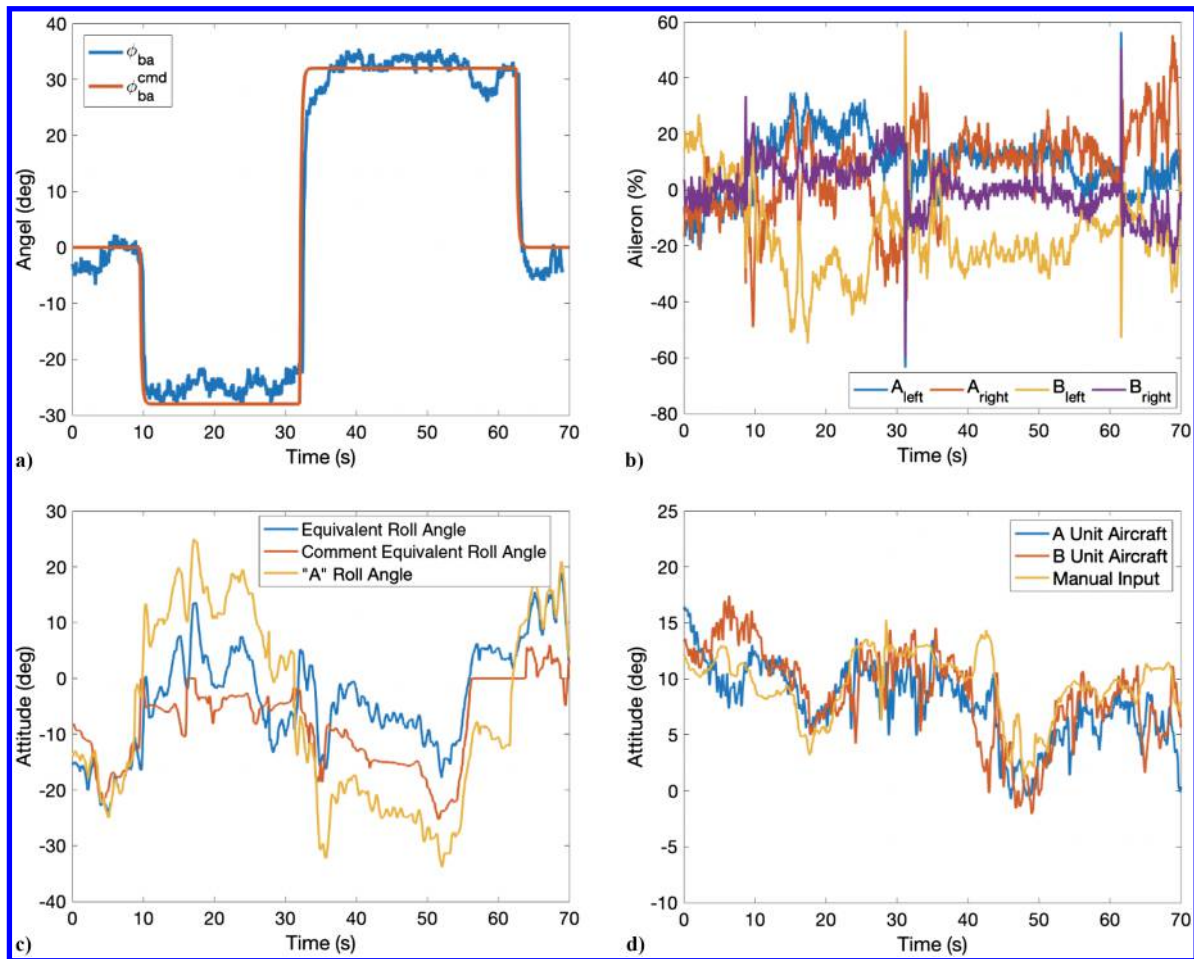


Fig. 13 Data on the morphing experiment.

A multibody dynamic model and a comprehensive control method, including a morphing controller and an overall attitude controller, have been developed. This approach enables the Compound Flexible Multibody Aircraft (CFMA) to collaboratively execute morphing

motions and overall attitude maneuvers. The system successfully maintains the desired configurations and switches between them seamlessly, as demonstrated in examinations involving CFMA composed of two and three unit aircraft.

The technology discussed in this paper facilitates the formation of a composite system of multiple small-aspect-ratio aircraft through wingtip hinges, significantly enhancing their flight efficiency while eliminating inherent instabilities. In the future, our efforts will focus on investigating which layouts of unit aircraft benefit most from this composite approach. Additionally, we aim to improve the manufacturing of unit aircraft and extend the control methods discussed in this paper to CFMA with a growing number of connections.

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Associate Editor